Canonical Positive Symplectic Diagonalization of the Pais–Uhlenbeck Oscillator

Abstract

We prove that the unequal-frequency PT-symmetric Pais–Uhlenbeck oscillator admits a canonical positive complex-symplectic diagonalization, and that the complexified-metaplectic logarithm of the unique Hermitian positive-definite diagonalizer is exactly the Bender–Mannheim generator $Q = \alpha pq + \beta xy$. The generator is thereby *reconstructed* from the normal-form data (G, J, G_0) rather than assumed. We determine the full solution family of the diagonalization problem: its stabilizer is $SO(2,\mathbb{C}) \times SO(2,\mathbb{C})$, of which generically only a finite subgroup is unitary in the canonical coordinates — and never the full maximal compact $U(1)^2$ — so the positive polar factor is not an invariant of the problem, while the Hermitian positive-definite diagonalizer is unique relative to the fixed normalized data. In canonical coordinates Q is a complexified beam-splitter generator with pure-point spectrum $r\,\mathbb{Z}$. We classify the algebraic positive metric forms compatible with pseudo-Hermiticity on a common invariant domain, compute the observable-space metric spectrum $\{e^r, e^r, e^{-r}, e^{-r}\}$ with $r = \log \frac{\omega_1+\omega_2}{\omega_1-\omega_2}$, and show that its affine-invariant distance from the identity is exactly $2r$; geometrically, the selected diagonalizer is the minimum-displacement representative of a non-θ-compatible reductive coset, with Cartan projection on the C_2 Weyl-chamber wall. In the equal-frequency limit this distance diverges logarithmically — escape to infinity in the symmetric space — and we prove that the limiting Jordan block admits no positive-definite invariant Hermitian form. All finite-dimensional algebraic identities appearing in this paper have been independently verified symbolically, numerically, and by machine-checked Lean 4 proofs.

1 Introduction

The Pais–Uhlenbeck (PU) oscillator [1] is the minimal higher-derivative model: a single degree of freedom governed by the fourth-order Lagrangian

$$L = \frac{\gamma}{2}\left(\ddot{z}^{\,2} - (\omega_1^2+\omega_2^2)\,\dot{z}^{\,2} + \omega_1^2\omega_2^2\,z^2\right), \qquad \gamma > 0, \quad \omega_1 > \omega_2 > 0. \tag{1}$$

Its Ostrogradsky Hamiltonian is unbounded below, and canonical quantization on the naive inner product produces negative-norm states. This ghost problem is the standard obstruction to higher-derivative theories, including conformal gravity, where the same mode structure appears [7].

Bender and Mannheim [3, 4] observed that the Ostrogradsky Hamiltonian, continued to a PT-symmetric operator H_{PT}, has real positive spectrum, and that an operator $Q = \alpha pq + \beta xy$ with explicit coefficients intertwines H_{PT} with a manifestly positive two-oscillator Hamiltonian: $e^{-Q/2} H_{\mathrm{PT}}\, e^{Q/2}$ is Hermitian, and $\eta = e^{-Q}$ is a positive metric restoring unitarity. Their derivation proceeds by an operator ansatz whose coefficients are fixed by demanding that the transformed Hamiltonian be Hermitian. Related constructions eliminate the PU ghost by complex canonical transformations [12] or by an alternative Hamiltonian formulation [11]; the general framework of pseudo-Hermitian quantum mechanics is due to Scholtz, Geyer and Hahne [8] and Mostafazadeh [9, 10].

This paper treats the problem as one of *finite-dimensional symplectic geometry* and proves the following.

1. The PT Hamiltonian is a complex quadratic form $H_{\mathrm{PT}} = \frac{1}{2}V^\top GV$ with a complex-symmetric (not Hermitian) matrix G, and the diagonalization problem
$$S^\top JS = J, \qquad S^\top GS = G_0 \tag{2}$$
with G_0 the positive two-oscillator normal form, has an explicit solution family.

2. Given the ordered target normal form G_0, the fixed canonical (Euclidean) adjoint, and the canonical symplectic normalization (Section 3), the problem (2) has a *unique* Hermitian positive-definite solution $S_+ \in \mathrm{Sp}(4,\mathbb{C})$. Uniqueness requires no genericity assumption beyond $\omega_1 > \omega_2 > 0$; it is uniqueness relative to these fixed normalized data, and the corresponding coordinate-free statement is an equivalence class under the identified canonical rescalings (Remark 3.6).

3. The metaplectic logarithm of S_+ is exactly the Bender–Mannheim generator: the quadratic operator $\widehat{Q}$ with $e^{-\widehat{Q}/2}\, V\, e^{\widehat{Q}/2} = S_+V$ is $\widehat{Q} = \alpha pq + \beta xy$ with the Bender–Mannheim coefficients. The generator is thus reconstructed from (G, J, G_0) and positivity alone (Section 4); no ansatz enters.

4. The stabilizer of the normal form is $SO(2,\mathbb{C}) \times SO(2,\mathbb{C})$ (Section 3). For a generic normalized Euclidean structure only the finite subgroup $\mathbb{Z}_2 \times \mathbb{Z}_2$ of it is unitary in the canonical coordinates (one $U(1)$ factor appears on an exceptional aspect-alignment locus, never the full $U(1)^2$), and consequently the positive polar factor of a general solution of (2) is *not* unique; the correct uniqueness statement concerns the Hermitian positive-definite solution.

5. The algebraic positive metric forms η' compatible with pseudo-Hermiticity of H_{PT}, on a specified common invariant domain, form the family $\eta' = \rho^\dagger W \rho$ with $W > 0$ commuting with the diagonalized Hamiltonian; positive metrics are not unique (Section 5).

6. The observable-space metric $M_{\mathrm{obs}} = S_+^\dagger S_+$ has spectrum $\{e^r, e^r, e^{-r}, e^{-r}\}$ with $r = \log \frac{\omega_1+\omega_2}{\omega_1-\omega_2}$, and its affine-invariant distance from the identity is exactly
$$d(I, M_{\mathrm{obs}}) = 2r = 2\log\frac{\omega_1+\omega_2}{\omega_1-\omega_2} \tag{3}$$
(Section 6). For $\omega_{1,2} = \omega \pm \epsilon$ one has $r = \log(\omega/\epsilon)$ exactly, so the equal-frequency exceptional point lies at infinite affine-invariant distance, and no scalar renormalization of M_{obs} has a finite positive nondegenerate limit.

7. At the exceptional point the flow develops a Jordan block, and we prove that a Jordan block admits no positive-definite invariant Hermitian form: nondegenerate invariant forms are indefinite and positive-semidefinite ones are degenerate (Section 7).

Throughout, we distinguish the observable-space (phase-space) geometry of the 4×4 matrices from the Hilbert-space geometry of the unbounded metric operator $\eta = e^{-Q}$; the two are related by the metaplectic representation but must not be identified (Remark 6.4).

Verification. All finite-dimensional algebraic identities appearing in this paper have been independently verified symbolically (SymPy), numerically (50–80 significant digits, including a near-degenerate frequency pair), and—for the identities $K^2 = -r^2 I$, $S^\top J S = J$, $\det S = 1$, $S^\top G S = G_0$, and the complete Jordan-block obstruction of Section 7—by machine-checked Lean 4 proofs against Mathlib, with no unproved placeholders. The accompanying repository contains the complete verification pipeline. Operator-theoretic statements about the unbounded metric η are made at the formal algebraic level on the oscillator domain and are flagged as such.

2 Quadratic Hamiltonian formulation

2.1 Conventions

We collect the canonical operators in the column vector $V = (x, y, p, q)^\top$ with

$$[x, p] = i, \qquad [y, q] = i, \tag{4}$$

all other basic commutators vanishing; equivalently $[V_a, V_b] = iJ_{ab}$ with the symplectic matrix

$$J = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \qquad J^\top = -J, \qquad J^2 = -I. \tag{5}$$

A quadratic Hamiltonian $H = \frac{1}{2} V^\top G V$ with $G = G^\top$ generates the linear flow $\dot{V} = AV$, $A = JG$. The same formulas define the classical theory with $\{V_a, V_b\} = J_{ab}$; all matrix identities below are representation-independent.

2.2 The PT Hamiltonian

Introducing the Ostrogradsky variables $x = \dot{z}$ and their momenta, the Lagrangian (1) gives $H_{\mathrm{PU}} = p^2/(2\gamma) + p_z x + \frac{\gamma}{2}(\omega_1^2 + \omega_2^2)x^2 - \frac{\gamma}{2}\omega_1^2\omega_2^2 z^2$, whose $-z^2$ term encodes the Ostrogradsky instability. The PT reformulation [3, 4] substitutes $z = iy$, $p_z = -iq$ (preserving $[y, q] = i$) and yields

$$H_{\mathrm{PT}} = \frac{p^2}{2\gamma} - iqx + \frac{\gamma}{2}(\omega_1^2 + \omega_2^2)x^2 + \frac{\gamma}{2}\omega_1^2\omega_2^2\, y^2 = \frac{1}{2} V^\top G V, \tag{6}$$

$$G = \begin{pmatrix} \gamma(\omega_1^2 + \omega_2^2) & 0 & 0 & -i \\ 0 & \gamma\omega_1^2\omega_2^2 & 0 & 0 \\ 0 & 0 & \gamma^{-1} & 0 \\ -i & 0 & 0 & 0 \end{pmatrix}, \tag{7}$$

where the cross term uses $[x, q] = 0$, so no ordering ambiguity arises. The matrix G is complex symmetric but not Hermitian: H_{PT} is PT-symmetric, not Hermitian.

Proposition 2.1. *The flow matrix $A = JG$ has characteristic polynomial*

$$\det(\lambda I - A) = (\lambda^2 + \omega_1^2)(\lambda^2 + \omega_2^2), \tag{8}$$

computed directly from the determinant. For $\omega_1 \neq \omega_2$ the eigenvalues $\pm i\omega_1, \pm i\omega_2$ are distinct and purely imaginary.

The target normal form is the positive two-oscillator Hamiltonian

$$\widetilde{H} = \frac{p^2}{2\gamma} + \frac{q^2}{2\gamma\omega_1^2} + \frac{\gamma\omega_1^2}{2}x^2 + \frac{\gamma\omega_1^2\omega_2^2}{2}y^2 = \frac{1}{2}V^\top G_0 V, \qquad G_0 = \operatorname{diag}\bigl(\gamma\omega_1^2,\ \gamma\omega_1^2\omega_2^2,\ \gamma^{-1},\ (\gamma\omega_1^2)^{-1}\bigr)\,, \tag{9}$$

whose modes (x, p) and (y, q) oscillate with frequencies ω_1 and ω_2 respectively. Since G is complex, the classical Williamson theorem [18] for real positive forms does not apply; the content of Sections 3–4 is a positive symplectic normal form over $\mathbb{C}$ for this PT class, together with a uniqueness theorem.

3 Canonical positive symplectic diagonalization

3.1 The parameters

Throughout, set

$$r = \log\frac{\omega_1 + \omega_2}{\omega_1 - \omega_2} > 0, \qquad \sigma = \sqrt{\omega_1^2 - \omega_2^2} > 0. \tag{10}$$

Exponentiating $r/2$ gives the hyperbolic values used repeatedly below:

$$\cosh\frac{r}{2} = \frac{\omega_1}{\sigma}, \qquad \sinh\frac{r}{2} = \frac{\omega_2}{\sigma}, \qquad \cosh r = \frac{\omega_1^2 + \omega_2^2}{\omega_1^2 - \omega_2^2}, \qquad \sinh r = \frac{2\omega_1\omega_2}{\omega_1^2 - \omega_2^2}. \tag{11}$$

3.2 Canonical normalization

For $d_x, d_y > 0$ the diagonal matrix $D = \operatorname{diag}(d_x, d_y, d_x^{-1}, d_y^{-1})$ satisfies $D^\top J D = J$: it is a real canonical rescaling. We normalize the coordinates by

$$V' = DV, \qquad d_x d_y = \gamma\omega_1\omega_2, \tag{12}$$

and write $G' = D^{-\top} G D^{-1}$, $G_0' = D^{-\top} G_0 D^{-1}$ for the quadratic forms in the normalized coordinates. Only the product $d_x d_y$ is fixed; every statement below is invariant under the residual split d_x/d_y, which acts by conjugation with a positive diagonal symplectic matrix. For the symmetric choice $d_x = d_y = \sqrt{\gamma\omega_1\omega_2}$,

$$G' = \begin{pmatrix} \frac{\omega_1^2+\omega_2^2}{\omega_1\omega_2} & 0 & 0 & -i \\ 0 & \omega_1\omega_2 & 0 & 0 \\ 0 & 0 & \omega_1\omega_2 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix}, \qquad G_0' = \operatorname{diag}\bigl(t,\ \omega_1\omega_2,\ \omega_1\omega_2,\ t^{-1}\bigr)\,, \qquad t = \frac{\omega_1}{\omega_2}. \tag{13}$$

Both G' and G_0' are block diagonal with respect to the planes $\Pi_{xq} = \operatorname{span}\{x, q\}$ and $\Pi_{yp} = \operatorname{span}\{y, p\}$.

The reason for the condition $d_x d_y = \gamma\omega_1\omega_2$ is Proposition 3.3 below: it is exactly the normalization that renders the diagonalizer Hermitian.

3.3 Existence

Definition 3.1. Let $B \in M_4(\mathbb{C})$ be given, in the ordering (x, y, p, q), by

$$B = \begin{pmatrix} 0 & 0 & 0 & i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix}, \qquad B^\dagger = B, \qquad B^2 = I, \tag{14}$$

a Hermitian involution with two-dimensional eigenspaces $\ker(B \mp I)$, and set

$$S_+ = \cosh\frac{r}{2}\, I + \sinh\frac{r}{2}\, B = \exp\Big(\frac{r}{2}B\Big) = \frac{tI + B}{\sqrt{t^2 - 1}}. \tag{15}$$

The last equality follows from (11). Since B is a Hermitian involution, S_+ is Hermitian with spectrum $\{e^{r/2}, e^{r/2}, e^{-r/2}, e^{-r/2}\}$; in particular $S_+ > 0$ and $\det S_+ = 1$.

Theorem 3.2 (Existence)**.** *The matrix S_+ of (15) satisfies*

$$S_+^\top J S_+ = J, \qquad S_+^\top G' S_+ = G_0'. \tag{16}$$

Proof. $B^\top J = -JB$ by inspection, so B is J-Hamiltonian and $e^{(r/2)B}$ is symplectic: $S_+^\top J S_+ = (\cosh^2\frac{r}{2} - \sinh^2\frac{r}{2})J + \cosh\frac{r}{2}\sinh\frac{r}{2}\,(B^\top J + JB) = J$, using $B^\top JB = -JB^2 = -J$.

For the congruence, note that S_+ preserves the splitting $\Pi_{xq} \oplus \Pi_{yp}$, acting on each plane (in the bases (x, q) and (y, p), with $c = \cosh\frac{r}{2}$, $s = \sinh\frac{r}{2}$) by $\left(\begin{smallmatrix} c & is \\ -is & c \end{smallmatrix}\right)$, while by (13) G' restricts to $\left(\begin{smallmatrix} t+t^{-1} & -i \\ -i & 0 \end{smallmatrix}\right)$ on Π_{xq} and to $\omega_1\omega_2 I_2$ on Π_{yp}. On Π_{yp}, $\left(\begin{smallmatrix} c & -is \\ is & c \end{smallmatrix}\right)\left(\begin{smallmatrix} c & is \\ -is & c \end{smallmatrix}\right) = (c^2 - s^2)I_2 = I_2$, matching $G_0'|_{\Pi_{yp}} = \omega_1\omega_2 I_2$. On Π_{xq}, using $c = \omega_1/\sigma$, $s = \omega_2/\sigma$,

$$\begin{pmatrix} c & -is \\ is & c \end{pmatrix}\begin{pmatrix} t+t^{-1} & -i \\ -i & 0 \end{pmatrix}\begin{pmatrix} c & is \\ -is & c \end{pmatrix} = \begin{pmatrix} (t+t^{-1})c^2 - 2cs & i\big((t+t^{-1})cs - c^2 - s^2\big) \\ -i\big((t+t^{-1})cs - c^2 - s^2\big) & -(t+t^{-1})s^2 + 2cs \end{pmatrix} = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix},$$

where the off-diagonal entries vanish because $(t + t^{-1})cs = (\omega_1^2 + \omega_2^2)/\sigma^2 = c^2 + s^2$, and the diagonal entries reduce to ω_1/ω_2 and ω_2/ω_1 by direct simplification. This is G_0' restricted to the two planes. $\square$

Undoing the normalization: for any normalizer D_0 satisfying $d_x d_y = \gamma\omega_1\omega_2$, the matrix $S_{\text{orig}} = D_0^{-1} S_+ D_0$ depends only on the product (it is invariant under the residual split) and solves the original problem $S^\top JS = J$, $S^\top GS = G_0$; the congruence, symplecticity and $\det S = 1$ in the original coordinates are among the Lean-verified statements. The flow relation $S^{-1}AS = A_0 = JG_0$ follows from the congruence by $JS^\top = S^{-1}J$. Note that S_{orig} is a diagonalizer but is *not* Hermitian in the original coordinates (unless $\gamma\omega_1\omega_2 = 1$): Hermiticity singles out the normalized coordinates, as follows.

Proposition 3.3 (Hermiticity requires the normalization)**.** *Let $D = \operatorname{diag}(d_x, d_y, d_x^{-1}, d_y^{-1})$ with $d_x, d_y > 0$ be an arbitrary canonical rescaling $V' = DV$, and let*

$$S(D) = D\, S_{\text{orig}}\, D^{-1} = \cosh\tfrac{r}{2}\, I + \sinh\tfrac{r}{2}\, B(D), \qquad B(D) = D\,(D_0^{-1} B D_0)\, D^{-1}, \tag{17}$$

be the transported canonical solution of the D-transformed problem $(G'(D), J, G_0'(D))$, $G'(D) = D^{-\top} G D^{-1}$. Then $S(D)$ solves the transformed problem for every D, *and $S(D)$ is Hermitian if* *and only if* $d_x d_y = \gamma\omega_1\omega_2$*. (For* D *violating the normalization, the fixed normalized-coordinate formula* $c\,I + s\,B$ *is* not even a diagonalizer *of the transformed problem: at* $\gamma = 1$*,* $\omega_1 = 3$*,* $\omega_2 = 2$*,* $D = I$*, one finds* $(S_+^\top G S_+)_{11} = 21 \neq 9 = (G_0)_{11}$*. Being a solution of the congruence problem, being Hermitian, and choosing normalized coordinates are three distinct conditions; the proposition relates the last two along the solution family.)*

Proof. $S(D)$ solves the transformed problem because congruence problems transport covariantly under $V' = DV$ (machine-verified for symbolic D). Conjugation scales the entries: $B(D)_{xq}, B(D)_{yp}$ carry the factor $d_x d_y/(\gamma\omega_1\omega_2)$ and $B(D)_{py}, B(D)_{qx}$ its inverse; Hermiticity of the transported involution, hence of $S(D)$, holds iff the factor is 1. □

3.4 The solution family and uniqueness

We now determine *all* solutions of the diagonalization problem in the normalized coordinates. Write $A_0' = JG_0'$ and let P_1, P_2 be the coordinate projections onto the modes (x,p) and (y,q) of G_0', and

$$X_1 = \frac{P_1 A_0' P_1}{\omega_1}, \qquad X_2 = \frac{P_2 A_0' P_2}{\omega_2}, \qquad X_j^2 = -P_j. \tag{18}$$

Explicitly X_1 acts on (x,p) by $\left(\begin{smallmatrix} 0 & \lambda_1 \\ -\lambda_1^{-1} & 0 \end{smallmatrix}\right)$ and X_2 on (y,q) by $\left(\begin{smallmatrix} 0 & \lambda_2^{-1} \\ -\lambda_2 & 0 \end{smallmatrix}\right)$, where $\lambda_1, \lambda_2 > 0$ are the *dimensionless mode-aspect parameters* of the chosen canonical coordinates: with $\hbar = 1$ and the symmetric split $d_x = d_y$ they take the values $\lambda_1 = \omega_2$, $\lambda_2 = \omega_1$ in normalized units, and a residual split d_x/d_y rescales them individually while preserving the split-invariant ratio $\lambda_1/\lambda_2 = \omega_2/\omega_1 < 1$. All exceptional conditions below are alignments $\lambda_j = 1$ of a mode aspect with the canonical split — comparisons of dimensionless quantities, not of frequencies with the number 1.

Lemma 3.4 (Stabilizer)**.** *For* $\omega_1 \neq \omega_2$*,*

$$\mathrm{Stab}(J, G_0') = \{C :\ C^\top J C = J,\ C^\top G_0' C = G_0'\} = \left\{C(\theta_1,\theta_2) = \textstyle\sum_{j=1,2} \left(\cos\theta_j\, P_j + \sin\theta_j\, X_j\right) :\ \theta_j \in \mathbb{C}\right\} \ \cong\ SO(2 \tag{19}$$

Proof. $(\supseteq)$ On each mode, $X_j^\top J + JX_j = 0$ and $X_j^\top G_0' + G_0' X_j = 0$ by inspection, and $X_j^2 = -P_j$, so for $C_j = \cos\theta_j P_j + \sin\theta_j X_j$ both quadratic forms are preserved: $C_j^\top Q C_j = (\cos^2\theta_j + \sin^2\theta_j)Q = Q$ on the mode, for $Q \in \{J, G_0'\}$.

$(\subseteq)$ If C preserves both forms then, using $C^{-\top} = JCJ^{-1}$ from symplecticity, $A_0' C = JG_0' C = JC^{-\top} G_0' = CJG_0' = CA_0'$: C commutes with A_0'. Since A_0' has the four distinct eigenvalues $\pm i\omega_1, \pm i\omega_2$ (Proposition 2.1), its commutant is the span of $\{P_1, X_1, P_2, X_2\}$. Writing $C = \sum_j (a_j P_j + b_j X_j)$, the condition $C^\top J C = J$ restricts on each mode to $a_j^2 + b_j^2 = 1$, which is the $SO(2,\mathbb{C})$ relation; $C^\top G_0' C = G_0'$ is then automatic as in $(\supseteq)$. □

Consequently the full solution family of (2) in normalized coordinates is the coset

$$\{S :\ S^\top J S = J,\ S^\top G' S = G_0'\} = S_+ \cdot \mathrm{Stab}(J, G_0'). \tag{20}$$

(If S is any solution, $C = S_+^{-1} S$ preserves both forms.)

Theorem 3.5 (Uniqueness relative to the fixed data)**.** *Fix the ordered target normal form* G_0'*, the canonical (Euclidean) adjoint defining Hermiticity, and the normalization* (12)*. Then for every* $\gamma > 0$ *and* $\omega_1 > \omega_2 > 0$*, the matrix* S_+ *is the unique Hermitian positive-definite solution of* $S^\top J S = J$*,* $S^\top G' S = G_0'$*.*

Proof. By (20) any solution is $S_+C(\theta_1,\theta_2)$. Writing out the 16 entries of S_+C ($c=\cosh\frac{r}{2},\ s=\sinh\frac{r}{2}$), the Hermiticity conditions $(S_+C)^\dagger = S_+C$ include, from the entry pairs $(x,p)/(p,x)$ and $(y,q)/(q,y)$, in terms of the dimensionless mode aspects λ_j (equal to $\lambda_1=\omega_2$, $\lambda_2=\omega_1$ in normalized units at the symmetric split),

$$\lambda_1^2 \sin\theta_1 = -\overline{\sin\theta_1}, \qquad \sin\theta_2 = -\lambda_2^2\,\overline{\sin\theta_2}. \tag{21}$$

Taking absolute values, $\lambda_1^2\,|\sin\theta_1| = |\sin\theta_1|$ and $|\sin\theta_2| = \lambda_2^2\,|\sin\theta_2|$; hence $\sin\theta_1=0$ unless $\lambda_1=1$ and $\sin\theta_2=0$ unless $\lambda_2=1$, and these two exceptional alignments cannot occur simultaneously since $\lambda_1/\lambda_2=\omega_2/\omega_1<1$ is split-invariant. If $\lambda_1=1$, the (y,q) condition still gives $\sin\theta_2=0$, and the entry pair $(x,y)/(y,x)$ gives $\lambda_1\lambda_2\sin\theta_2 = -\overline{\sin\theta_1}$, forcing $\sin\theta_1=0$ as well; the case $\lambda_2=1$ is symmetric. So in all cases $\sin\theta_1=\sin\theta_2=0$, whence $\cos^2\theta_j=1$ and $C=\varepsilon_1P_1+\varepsilon_2P_2$ with $\varepsilon_j=\pm1$. The entry pair $(x,q)/(q,x)$ gives $\cos\theta_2=\overline{\cos\theta_1}$, i.e. $\varepsilon_1=\varepsilon_2$, so $C=\pm I$. Finally $S_+(-I)=-S_+<0$ is excluded by positivity, leaving $C=I$. Every exceptional condition in this proof is an alignment $\lambda_j=1$ of dimensionless quantities. ☐

Remark 3.6 (Intrinsic uniqueness versus canonical object). The theorem is a uniqueness statement *within fixed normalized data*: the ordered mode assignment (ω_1 to (x,p), ω_2 to (y,q)), the target aspect ratios in G_0', the Euclidean adjoint, the normalization $d_xd_y=\gamma\omega_1\omega_2$, and the variable ordering are all part of the input. The corresponding coordinate-independent object is the equivalence class of S_+ under the canonical rescalings identified in Proposition 3.3 (the residual split acts trivially) — not a canonical matrix attached to the bare pair (G,J).

Proposition 3.7 (Polar factors are not invariant)**.** *Let $S=S_+C$ with $C\in\mathrm{Stab}(J,G_0')$, and let $S=PU$ be its polar decomposition ($P>0$, U unitary). Then $P=S_+$ if and only if C is unitary. In the normalized coordinates the unitary elements of* $\mathrm{Stab}(J,G_0')$ *form, for $\lambda_1,\lambda_2\neq1$, the finite group $\{\pm P_1\pm P_2\}\cong\mathbb{Z}_2\times\mathbb{Z}_2$. Consequently the positive polar factor varies over a two-complex-parameter family as C ranges over the stabilizer: it is not an invariant of the diagonalization problem.*

Proof. If C is unitary, S_+C is already a polar decomposition, and by its uniqueness $P=S_+$. Conversely $P=S_+$ forces $U=S_+^{-1}S=C$ unitary. For the second claim, $C(\theta)$ restricted to mode 1 is $\begin{pmatrix} a & b\lambda_1 \\ -b/\lambda_1 & a\end{pmatrix}$ with $a^2+b^2=1$; unitarity requires $|a|^2+|b|^2\lambda_1^{\mp2}=1$ for both signs, hence $b=0$ for $\lambda_1\neq1$, and then $|a|=1$ with $a^2=1$, i.e. $a=\pm1$; likewise on mode 2. An explicit non-unitary example is $C(i\tau,0)$, $\tau\in\mathbb{R}\setminus\{0\}$, for which $(S_+C)^\dagger(S_+C)\neq S_+^2$. ☐

Remark 3.8 (Unitary part: generic, exceptional, and invariant statements). The stabilizer is $SO(2,\mathbb{C})^2\cong(\mathbb{C}^\times)^2$ as an abstract group, whose maximal compact subgroup is $U(1)^2$. Its unitary part in the canonical coordinates depends on the normalized Euclidean structure: for a *generic* structure ($\lambda_1,\lambda_2\neq1$) it is the finite group $\mathbb{Z}_2\times\mathbb{Z}_2$; on the exceptional aspect-alignment locus $\lambda_j=1$ (which the residual split freedom d_x/d_y of (12) can move but not remove — the ratio λ_1/λ_2 is split-invariant) it is enlarged by the one $U(1)$ factor of that mode. The *invariant* statement is that the unitary part can never contain the full maximal compact $U(1)^2$, because $\lambda_1/\lambda_2=\omega_2/\omega_1\neq1$: the stabilizer H is not compatible with the selected Cartan involution — its intersection with the maximal compact subgroup K is not a maximal compact subgroup of H — whatever the normalized Euclidean structure. Theorem 3.5 holds for all parameters.

4 Reconstruction of the Bender–Mannheim generator

We now derive the generator from S_+; at no point is the form of Q assumed.

4.1 Quadratic generators and the metaplectic dictionary

For a symmetric matrix $M = M^\top \in M_4(\mathbb{C})$ let $\widehat{Q} = \frac{1}{2}V^\top MV$ and $K = JM$. From $[V_a, V_b] = iJ_{ab}$,

$$[\widehat{Q}, V] = \tfrac{1}{2}M_{bc}\big(V_b[V_c, V_a] + [V_b, V_a]V_c\big)_a = -\,iKV, \tag{22}$$

componentwise on the column vector V; the sign is fixed by the convention $[x, p] = +i$. Since $\mathrm{ad}_{\widehat{Q}}$ acts linearly on V, the Baker–Campbell–Hausdorff series sums exactly:

$$e^{-\widehat{Q}/2}\, V\, e^{\widehat{Q}/2} = e^{iK/2}\, V, \qquad e^{\widehat{Q}/2}\, V\, e^{-\widehat{Q}/2} = e^{-iK/2}\, V. \tag{23}$$

Thus quadratic operators implement one-parameter symplectic groups: this is the (formal) metaplectic dictionary $\widehat{Q} \leftrightarrow K = JM$ between quadratic generators and the Lie algebra $\mathfrak{sp}(4, \mathbb{C})$ [23]. A terminological caution: the genuine metaplectic group is the double cover of the *real* symplectic group and acts unitarily; here $S_+ \in \mathrm{Sp}(4, \mathbb{C})$ is non-real and is implemented by an unbounded nonunitary operator. "Metaplectic" below always means this *complexified oscillator representation* — the exponentiated representation of $\mathfrak{sp}(4, \mathbb{C})$ on the algebraic oscillator domain — and "metaplectic logarithm" is used in this formal complex sense.

4.2 The logarithm of S_+

By Definition 3.1, $S_+ = e^{(r/2)B}$, and since B is an involution this is the exact logarithm delivered by the spectral calculus:

$$\log S_+ = \frac{r}{2}\, B. \tag{24}$$

To realize $S_+ = e^{iK'/2}$ in the dictionary (23) we need $iK'/2 = \log S_+$, i.e.

$$K' = -\,irB, \qquad M' = J^{-1}K' = -JK' = r\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad \text{(normalized coordinates)}, \tag{25}$$

which is symmetric, as required. Pulling back along the normalization (12), $M = D^\top M'D$ depends only on the product $d_x d_y = \gamma\omega_1\omega_2$ and equals

$$M = \begin{pmatrix} 0 & \beta & 0 & 0 \\ \beta & 0 & 0 & 0 \\ 0 & 0 & 0 & \alpha \\ 0 & 0 & \alpha & 0 \end{pmatrix}, \qquad \alpha = \frac{r}{\gamma\omega_1\omega_2}, \qquad \beta = r\,\gamma\omega_1\omega_2 = \alpha\gamma^2\omega_1^2\omega_2^2. \tag{26}$$

Theorem 4.1 (Reconstruction)**.** *The quadratic operator associated with the metaplectic logarithm of the unique Hermitian positive-definite diagonalizer is*

$$Q = \tfrac{1}{2}V^\top MV = \alpha\, pq + \beta\, xy, \qquad \alpha = \frac{1}{\gamma\omega_1\omega_2}\log\frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}, \qquad \beta = \alpha\gamma^2\omega_1^2\omega_2^2, \tag{27}$$

i.e. exactly the Bender–Mannheim generator [4], with $r = \sqrt{\alpha\beta}$ *(positive root, since* $\alpha, \beta > 0$ *for* $\omega_1 > \omega_2 > 0$*). Moreover* $K = JM$ *satisfies* $K^2 = -r^2 I$*, so the exponential closes in the first two powers of* K*,*

$$S = e^{iK/2} = \cosh\frac{r}{2}\, I + i\,\frac{\sinh\frac{r}{2}}{r}\, K, \tag{28}$$

and $e^{-Q/2} H_{\mathrm{PT}}\, e^{Q/2} = \frac{1}{2} V^\top (S^\top G S) V = \frac{1}{2} V^\top G_0 V = \widetilde{H}$ *is the positive two-oscillator Hamiltonian (9).*

Proof. The pullback computation above gives M, hence Q; the identities $K^2 = -r^2 I$ and $S^\top G S = G_0$ are verified directly (and are machine-checked); the last display combines (23) with the congruence. $\square$

The logical order of the construction is therefore

$$(G,\ J,\ G_0) \implies S_+ \text{ (existence and uniqueness)} \implies Q \text{ (metaplectic logarithm)}, \tag{29}$$

with positivity of S_+ (after the canonical normalization (12)) as the only selection principle. Beyond H_{PT} itself, the inputs are the choice of the target normal form G_0 and the positivity requirement; nothing else.

5 Pseudo-Hermiticity and metric classification

In this section, operator identities are formal-algebraic: they hold on the common invariant domain $\mathcal{F}_{\mathrm{alg}}$ of finite linear combinations of the $\widetilde{H}$-oscillator eigenstates, on which all operators below act. This domain is genuinely invariant: in the normalized coordinates $Q' = r(x'y' + p'q') = r(a'_1 a'^\dagger_2 + a'^\dagger_1 a'_2)$ is a complexified beam-splitter generator commuting with the total number operator (Proposition 5.1 below), so each finite-total-number sector is a finite-dimensional invariant subspace for Q, $e^{\pm Q/2}$, and all polynomial quadratic Hamiltonians. Q is essentially self-adjoint there but $e^{\pm Q/2}$ are unbounded, and we make no claims about closures or operator-theoretic similarity: the classification below is of *algebraic positive metric forms on* $\mathcal{F}_{\mathrm{alg}}$, not of closed operators. The finite-dimensional identities backing each step are exact and verified.

Since $\alpha, \beta \in \mathbb{R}$ and $[p, q] = [x, y] = 0$, the generator satisfies $Q^\dagger = Q$; set

$$\rho = e^{-Q/2} = \rho^\dagger, \qquad \eta = \rho^\dagger \rho = e^{-Q}. \tag{30}$$

Proposition 5.1 (Spectrum of Q)**.** *In the normalized coordinates, with* $a_1 = (x' + ip')/\sqrt{2}$*,* $a_2 = (y' + iq')/\sqrt{2}$*,*

$$Q' = r\,(x'y' + p'q') = r\left(a_1 a_2^\dagger + a_1^\dagger a_2\right) \tag{31}$$

exactly (the cross terms cancel because $[x, q] = [p, y] = 0$*; no ordering ambiguity arises).* Q' *commutes with* $N_{\mathrm{tot}} = N_1 + N_2$*, and on the fixed-*$N$ *sector (Schwinger* $SU(2)$*: the operator is* $2J_x$*) its spectrum is exactly* $r\{-N, -N+2, \ldots, N\}$*. Hence*

$$\operatorname{spec} Q = r\,\mathbb{Z} \quad \textit{(pure point, infinite multiplicities)}, \qquad \operatorname{spec}(e^{-Q}) = \{e^{-rn} : n \in \mathbb{Z}\} \cup \{0\}, \tag{32}$$

where 0 *is an accumulation point of the spectrum but not an eigenvalue:* η *is unbounded with unbounded inverse, but its spectrum is not* $(0, \infty)$*. Since the normalization map is a real* *symplectic rescaling, implemented unitarily, the same spectra hold in the original coordinates. The identity also gives $Q'\,\Omega_{\mathrm{can}} = 0$: the generator annihilates the canonical vacuum while carrying divergent Cartan data — the finite-dimensional shadow of which is Section 6.*

Proposition 5.2 (Pseudo-Hermiticity)**.** *Formally, $H_{\mathrm{PT}}^{\dagger}\,\eta = \eta\,H_{\mathrm{PT}}$. At the matrix level this is equivalent to the exact identity*

$$(S^2)^{\top}\,G\,S^2 = \overline{G}, \tag{33}$$

which holds for $S = e^{iK/2}$.

Proof. By (23) with parameter doubled, $e^{-Q}Ve^{Q} = S^2V$, so $\eta H_{\mathrm{PT}}\eta^{-1} = \frac{1}{2}V^{\top}\big((S^2)^{\top}GS^2\big)V$. Since x, y, p, q are Hermitian and G is symmetric, $H_{\mathrm{PT}}^{\dagger} = \frac{1}{2}V^{\top}\overline{G}V$. The claim is then (33), verified exactly. Equivalently: with $h = \rho H_{\mathrm{PT}}\rho^{-1} = \widetilde{H}$ Hermitian (Theorem 4.1) and $\rho^{\dagger} = \rho$, the free-algebra identity $\rho\,(h^{\dagger} - h)\,\rho = H_{\mathrm{PT}}^{\dagger}\eta - \eta H_{\mathrm{PT}}$ gives the statement. □

Theorem 5.3 (Classification of algebraic positive metric forms)**.** *Let $h = \rho H_{\mathrm{PT}}\rho^{-1} = \widetilde{H}$, all operators acting on the common invariant domain $\mathcal{F}_{\mathrm{alg}}$. The map $W \mapsto \eta' = \rho^{\dagger}W\rho$ is a bijection from*

$$\{W > 0 \text{ invertible on } \mathcal{F}_{\mathrm{alg}},\ [W,h] = 0\} \quad \text{onto} \quad \{\eta' > 0 \text{ invertible on } \mathcal{F}_{\mathrm{alg}},\ H_{\mathrm{PT}}^{\dagger}\eta' = \eta' H_{\mathrm{PT}}\}, \tag{34}$$

with inverse $\eta' \mapsto W = \rho^{-\dagger}\eta'\rho^{-1}$, as an identity of quadratic forms on $\mathcal{F}_{\mathrm{alg}}$. In particular positive metric forms are not *unique: $W = I$ gives the canonical $\eta = e^{-Q}$, and any positive invertible function $W = f(N_1, N_2)$ of the two commuting number operators of $\widetilde{H}$ gives another. For ω_1/ω_2 irrational the bounded commutant of h is generated by bounded functions of N_1, N_2 (for rational ratio the commutant is enlarged by energy degeneracies), so within operators affiliated with the joint spectral algebra of (N_1, N_2) this family is exhaustive. Closability, self-adjointness of the products, and operator-theoretic (as opposed to form) identities are not claimed and remain open at the unbounded level.*

Proof. With $\rho^{\dagger} = \rho$ and $H_{\mathrm{PT}} = \rho^{-1}h\rho$, $H_{\mathrm{PT}}^{\dagger} = \rho\,h\,\rho^{-1}$ (using $h^{\dagger} = h$), a two-line computation gives $H_{\mathrm{PT}}^{\dagger}\eta' - \eta' H_{\mathrm{PT}} = \rho\,[h, W]\,\rho$ for $\eta' = \rho W\rho$: the intertwining relation holds iff $[W, h] = 0$. Positivity and invertibility transfer both ways because congruence by the invertible ρ preserves them, and the two maps are mutually inverse by construction. □

Uniqueness of the metric therefore requires a selection principle beyond positivity and pseudo-Hermiticity. The results of Section 3 identify one: within the symplectic-Gaussian class, demanding that the metric arise from a Hermitian positive-definite *symplectic* diagonalizer selects $\eta = e^{-Q}$ uniquely (Theorem 3.5). The relation between the two freedoms must be stated with care: only the Gaussian (quadratic-exponential) subset of the W-family corresponds to metaplectic implementers of stabilizer elements (20); general positive functions $f(N_1, N_2)$ are non-Gaussian and have no 4×4 symplectic representation. The finite-dimensional stabilizer is the *quadratic/Gaussian shadow* of the full commutant metric freedom,

$$\text{Gaussian stabilizer freedom} \ \subsetneq\ \text{full commutant metric freedom}, \tag{35}$$

and the non-invariance of the polar factor (Proposition 3.7) is the finite-dimensional shadow of the Gaussian part.

6 Geometry of the positive metric cone

6.1 The observable-space metric

Definition 6.1. The observable-space metric of the diagonalization is $M_{\mathrm{obs}} = S_+^\dagger S_+ \in M_4(\mathbb{C})$, a positive-definite Hermitian matrix on the normalized phase-space coordinates.

Since S_+ is Hermitian, $M_{\mathrm{obs}} = S_+^2 = e^{rB} = \cosh r\, I + \sinh r\, B$, and B is an involution with two-dimensional eigenspaces, so with (11):

Theorem 6.2. $\operatorname{spec} M_{\mathrm{obs}} = \{e^r, e^r, e^{-r}, e^{-r}\}$, *i.e.*

$$M_{\mathrm{obs}} = e^r\, \Pi_+ + e^{-r}\, \Pi_-, \qquad \Pi_\pm = \tfrac{1}{2}(I \pm B). \tag{36}$$

This holds in the canonically normalized coordinates (12)*; in the original coordinates $S^\dagger S$ has a different, γ-dependent spectrum.*

On the cone of positive-definite Hermitian matrices we use the affine-invariant Riemannian metric $\langle U, V\rangle_G = \operatorname{tr}(G^{-1}UG^{-1}V)$ (no normalization factor), whose geodesic distance from the identity is $d(I, M) = \|\log M\|_F$ [19].

Theorem 6.3 (Distance)**.**

$$d(I, M_{\mathrm{obs}}) = \|\log M_{\mathrm{obs}}\|_F = \|rB\|_F = 2r = 2\log\frac{\omega_1+\omega_2}{\omega_1-\omega_2}. \tag{37}$$

Proof. $\log M_{\mathrm{obs}} = rB$ by the spectral calculus, and $\|rB\|_F = r\sqrt{\operatorname{tr} B^\dagger B} = r\sqrt{\operatorname{tr} I} = 2r$. ☐

Remark 6.4 (Observable space versus Hilbert space)*.* $M_{\mathrm{obs}} = S_+^\dagger S_+$ and $\eta = e^{-Q}$ must not be identified as operators. M_{obs} is a 4×4 matrix acting on phase-space labels: it measures how the canonical coordinates are deformed by the diagonalization, and its spectrum $\{e^{\pm r}\}$ is the squeezing content of S_+. The metric $\eta = e^{-Q}$ is an unbounded positive operator on Hilbert space with pure-point spectrum $\{e^{-rn} : n \in \mathbb{Z}\} \cup \{0\}$ (Proposition 5.1: Q is a complexified beam-splitter generator with $\operatorname{spec} Q = r\,\mathbb{Z}$, infinite multiplicities — not a dilation generator with continuous spectrum). They are related, not equal: $\rho = e^{-Q/2}$ is the formal metaplectic implementer of S_+ via (23), and $\eta = \rho^\dagger\rho$ is the Hilbert-space pullback of the standard inner product along ρ, while $M_{\mathrm{obs}} = S_+^\dagger S_+$ is the finite-dimensional pullback along S_+. The quantity that passes between the two levels is the rapidity r — appearing as the squeezing exponent of M_{obs} on phase space and as the eigenvalue spacing of Q on Hilbert space.

6.2 Cartan geometry of the selected diagonalizer

The distance theorem has a group-geometric upgrade, established in the companion paper [5] and stated here in the present notation because it supplies the invariant formulation of Section 6.

Convention. We use the Gram–Cartan projection $\mu(S) = \log\operatorname{spec}(S^\dagger S)$ restricted to the closed Weyl chamber — i.e. *twice* the standard singular-value projection $\mu_{\mathrm{std}}(S) = \log s(S)$. The singular values of S_+ are $e^{\pm r/2}$ (each twice), so

$$\mu(S_+) = (r, r), \qquad \mu_{\mathrm{std}}(S_+) = (r/2,\, r/2); \tag{38}$$

the Gram convention is the one synchronized with $d(I, M_{\text{obs}}) = \|\log S_+^\dagger S_+\|_F = 2r$ and with the distortion dictionary $F(S) = \|\log S^\dagger S\|_F^2 = 2\|\mu(S)\|^2$ of [5]; the two conventions must not be mixed.

Theorem 6.5 (Cartan-coset formulation [5])**.** *(i) The admissible diagonalizers form the coset S_+H, $H = \mathrm{Stab}(J, G_0') \cong SO(2,\mathbb{C})^2$ (20). (ii) The distortion functional $F(S) = \|\log S^\dagger S\|_F^2$ is squared symmetric-space displacement in* $\mathrm{Sp}(4,\mathbb{C})/\mathrm{Sp}(2)$*: $F(S) = 2\|\mu(S)\|^2$. (iii) S_+ minimizes F on the coset S_+H. (iv) $\mu(S_+) = (r, r)$ has equal components: the selected representative lies on the C_2 Weyl-chamber wall. (v) The minimization is* not *a direct application of the standard Mostow/Kempf–Ness nearest-point theorem to H, because H is not compatible with the canonical Cartan involution (Remark 3.8: $H \cap K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ is not a maximal compact subgroup of H); the proof proceeds through the θ-compatible supergroup $C \cong SL(2,\mathbb{C}) \times SL(2,\mathbb{C})$ containing H, whose orbit is totally geodesic.*

This turns the explicit matrix diagonalization into a geometric theorem about a non-θ-compatible reductive coset, and it reinterprets the equal-frequency divergence below: $\|\mu(S_+)\| \to \infty$ means the selected representative *escapes to infinity in the symmetric space, along the chamber wall*, as the dynamical orbit approaches the Jordan stratum.

6.3 The equal-frequency limit

Let $\omega_1 = \omega + \epsilon$, $\omega_2 = \omega - \epsilon$ with $0 < \epsilon < \omega$. Then (10) gives, *exactly*,

$$r = \log \frac{2\omega}{2\epsilon} = \log \frac{\omega}{\epsilon}, \qquad d(I, M_{\text{obs}}) = 2 \log \frac{\omega}{\epsilon} \xrightarrow[\epsilon \to 0^+]{} \infty. \tag{39}$$

Moreover $\sigma^2 = 4\omega\epsilon$ and (11) give the exact spectral form

$$S_+ = \sqrt{\frac{\omega}{\epsilon}}\, \Pi_+ + \sqrt{\frac{\epsilon}{\omega}}\, \Pi_-, \tag{40}$$

so the diagonalizer diverges like $\epsilon^{-1/2}$ while collapsing onto the rank-two spectral projector Π_+; the similarity transformation degenerates precisely at the exceptional point $\omega_1 = \omega_2$, where Proposition 2.1 shows the flow eigenvalues merge and A acquires Jordan blocks.

Proposition 6.6 (No scalar renormalization)**.** *For every function $c(\epsilon) > 0$, the spectrum of $c\, M_{\text{obs}}$ is $\{c\,\omega/\epsilon\ (\times 2),\ c\,\epsilon/\omega\ (\times 2)\}$, with condition number $(\omega/\epsilon)^2 \to \infty$. Hence $c\, M_{\text{obs}}$ has no finite, positive-definite, nondegenerate limit as $\epsilon \to 0^+$: keeping the top eigenvalue finite forces the bottom one to zero.*

The exceptional point of the PU oscillator therefore lies at infinite affine-invariant distance from the identity in the metric cone, and the divergence rate is universal: $d = 2\log(\omega/\epsilon)$ with no model-dependent prefactor. This is the geometric expression of the known breakdown of the Bender–Mannheim construction at equal frequencies, where $\alpha, \beta \to \infty$ [4].

7 The Jordan obstruction

At $\omega_1 = \omega_2$ the flow is no longer diagonalizable, and the question becomes whether *any* positive metric can exist for the limiting dynamics.

Lemma 7.1 (From the PU flow to the Jordan matrix)**.** *At $\omega_1 = \omega_2 = \omega$ the flow matrix $A = JG$ has eigenvalues $\pm i\omega$, each with a nontrivial Jordan block. On the generalized eigenspace of $+i\omega$ set $H = iA$ (restricted there): H has the real eigenvalue $-\omega$ with the same Jordan structure, and an invariant Hermitian form for the flow, $A^\dagger\eta + \eta A = 0$, restricts to the invariance relation $H^\dagger\eta = \eta H$ used below. The obstruction theorem applied to each Jordan chain of H therefore governs the PU flow itself.*

The following finite-dimensional theorem then answers the question negatively. Parts (1)–(3) for $n = 2$ are fully machine-checked in Lean. Before the explicit classification, note the one-line general argument: if $\eta > 0$ and $H^\dagger\eta = \eta H$, then $\eta^{1/2}H\eta^{-1/2}$ is Hermitian, hence diagonalizable, so H is diagonalizable and a nontrivial Jordan block is impossible. The explicit Hankel classification below is retained because it also classifies the degenerate and indefinite invariant forms.

Theorem 7.2 (Jordan no-go)**.** *Let $\omega \in \mathbb{R}$ and $H_J = \left(\begin{smallmatrix}\omega & 1\\ 0 & \omega\end{smallmatrix}\right)$, and let $\eta = \eta^\dagger \in M_2(\mathbb{C})$ satisfy the invariance relation $H_J^\dagger\eta = \eta H_J$. Then*

$$\eta = \begin{pmatrix} 0 & b \\ b & d \end{pmatrix}, \qquad b, d \in \mathbb{R}, \tag{41}$$

and consequently:

1. *no positive-definite invariant Hermitian form exists ($\eta_{11} = \langle e_1, \eta e_1\rangle = 0$);*
2. *every positive-semidefinite invariant form is degenerate ($\det\eta = -b^2 \geq 0$ forces $b = 0$);*
3. *every nondegenerate invariant form is indefinite: for $b \neq 0$ and $\varepsilon = (1+|d|)^{-1}$ the vectors $x_\pm = (1, \pm\varepsilon b)^\top$ satisfy $\langle x_+, \eta x_+\rangle > 0 > \langle x_-, \eta x_-\rangle$.*

For the $n\times n$ Jordan block, every invariant Hermitian form is anti-triangular Hankel—constant on anti-diagonals and vanishing above the main anti-diagonal—so $\eta_{11} = 0$ and the same trichotomy holds for all $n \geq 2$.

Proof. Writing $H_J = \omega I + N$ with N the nilpotent shift and ω real, invariance reduces to $N^\top\eta = \eta N$, i.e. $\eta_{j-1,k} = \eta_{j,k-1}$ with the boundary convention $\eta_{0,k} = 0$: η is Hankel with $\eta_{1,k} = 0$ for $k \leq n-1$. For $n = 2$ this gives $\eta_{11} = 0$ and $\eta_{12} = \eta_{21}$, which with Hermiticity makes $b = \eta_{12}$ real. (1) follows from $\eta_{11} = 0$; (2) from $\det\eta = -b^2$ together with positive semidefiniteness of the principal minors; (3) from $\langle x_\pm, \eta x_\pm\rangle = \varepsilon b^2(\pm 2 + \varepsilon d)$ and $\varepsilon|d| < 1$. The general-n statement follows from the Hankel structure and $\langle e_1, \eta e_1\rangle = \eta_{11} = 0$. □

This is consistent with the general theory of invariant Hermitian forms for matrices with Jordan structure [20] and with the physics of exceptional points [21, 22]: at $\omega_1 = \omega_2$ the PU oscillator admits no positive nondegenerate invariant metric, degenerate or indefinite forms being the only invariant structures. Together with Proposition 6.6, the equal-frequency limit is seen from inside the metric cone as a boundary point at infinite distance, and from the limiting dynamics as an *empty cone of nondegenerate positive-definite invariant forms* — positive-semidefinite invariant forms exist but are degenerate (part 2), and such singular forms can appear naturally in exceptional-point limits.

Remark 7.3 (What the no-go does and does not exclude). The theorem excludes a nondegenerate positive-definite invariant metric at the Jordan point; it does *not* exclude equal-frequency PT-symmetric realizations as such. The precise conclusions are: the unequal-frequency positive metric has no nondegenerate positive limit (Proposition 6.6); the Jordan dynamics is not similar to a Hermitian diagonalizable Hamiltonian (the one-line argument above); nondegenerate invariant forms are necessarily indefinite (part 3); and intrinsically Jordan or singular PT constructions, such as the equal-frequency realization of Bender and Mannheim [4], are separate boundary constructions outside the scope of this classification. This distinction matters for the comparison with Krein quantization in the companion papers.

8 Discussion

Relation to Bender–Mannheim. The operator Q, its coefficients α, β, and the positive Hamiltonian $\widetilde{H}$ agree exactly with [3, 4]. What is added here is, first, the logical inversion: Q is not an ansatz whose coefficients are fixed by demanding Hermiticity of the transformed Hamiltonian, but the metaplectic logarithm of a diagonalizer that is unique once positivity and the symplectic structure are imposed (Theorems 3.5 and 4.1). Second, the precise positivity statement: the diagonalizer is Hermitian positive only after the canonical normalization $d_x d_y = \gamma\omega_1\omega_2$ (Proposition 3.3), a point that is easy to miss when working with the operator form alone. Third, the solution family, its stabilizer $SO(2,\mathbb{C})^2$, and the non-invariance of the polar factor (Proposition 3.7) delimit exactly how much of the construction is canonical.

Relation to complex canonical transformations and metric non-uniqueness. Déctor, Morales-Técotl, Urrutia and Vergara [12] remove the PU ghost by a complex canonical transformation to two ordinary oscillators, and Mostafazadeh [11] by an alternative Hamiltonian formulation; the transformation S_+ belongs to the same circle of ideas at the level of existence. The classification of Theorem 5.3 is the PU instance of the general non-uniqueness of metric operators in pseudo-Hermitian quantum mechanics [8, 10]; see [17] for a recent discussion of the physical consequences of metric freedom, and [15, 16] for the time-dependent Dyson-map [14] setting in which such maps are studied dynamically. The contribution of Sections 3–5 is to make both the freedom (the stabilizer, the W-family) and the canonical choice (unique positive symplectic diagonalizer) exact and finite-dimensional for this model.

Geometric interpretation. Within the normalization selected by Hermiticity of the positive diagonalizer, the affine distance is $2r = 2\log\frac{\omega_1+\omega_2}{\omega_1-\omega_2}$ and depends only on the frequency ratio: it is the geodesic length, in the natural geometry of the positive cone [19], separating the canonical observable-space metric from the identity. It is an invariant of the canonically normalized diagonalization problem $(G, J, G_0, \dagger)$ — not of the bare pair (G, J): the Euclidean adjoint, the reference identity, the target G_0, and the normalization all enter, and in the original coordinates $S^\dagger S$ has a different, γ-dependent spectrum (Theorem 6.2). The genuinely group-geometric formulation is the Cartan-coset theorem of Section 6.2: minimum symmetric-space displacement of a non-θ-compatible reductive coset, attained on the C_2 chamber wall. The exact logarithmic divergence, the $\epsilon^{-1/2}$ collapse of S_+ onto a rank-two projector, and the emptiness of the nondegenerate positive invariant cone at the Jordan point (Theorem 7.2) give the equal-frequency exceptional point a complete geometric characterization at the quadratic level: escape to infinity along a Weyl-chamber wall, ending at a Jordan stratum with no nondegenerate positive invariant form.

Sign of γ. Everything above assumes $\gamma > 0$ in (1). Applications to quadratic gravity use PU blocks with $\gamma < 0$ (the perfect-square convention in which the massless graviton is healthy and the massive spin-two branch is the ghost [6]); invoking the present theorems there requires an explicit transport — multiplication of the action by -1, an exchange of which normal mode is declared positive, a quarter-turn of one branch, or another specified real-form transformation. The matrix identities (normal form, stabilizer, logarithm, congruences) transport trivially under $G \to -G$, $G_0 \to -G_0$; what does *not* transport automatically is the positivity interpretation: the sign conventions of α, β, the positivity of the displayed target G_0, the positivity of Q, and the action-sign and commutator normalizations all flip or must be re-derived. The gravity companion [6] performs this transport explicitly.

Possible extensions. The construction used only the data (G, J, G_0) and positivity, and every step—normal form, stabilizer, logarithm, metric cone—is available for any finite set of PT-paired modes. The mode structure of fourth-order field theories, where each spatial momentum contributes a PU pair with $\epsilon(k)$ playing the role of the frequency splitting, suggests a possible extension; we do not pursue it here.

Verification companion. All finite-dimensional algebraic identities appearing in this paper have been independently verified symbolically, numerically, and by machine-checked Lean proofs. The artifact is the directory `physics/symplectic-reconstruction` of the accompanying repository: a symbolic audit of every identity (`symbolic/verify_sympy.py`, 51 claims, including the derivations of (22) and (33); `symbolic/verify_paper1_referee.py` covering the spectrum of Q on fixed-number sectors and the corrected normalization proposition, including the $D = I$ counterexample), a high-precision numerical regression suite covering generic and near-degenerate frequencies (`numeric/regression.py`, 50–80 digits), and a Lean 4 development against Mathlib v4.29.0 (`lean/PaisUhlenbeck/`, built with `lake build`, zero `sorry`) in which $K^2 = -r^2 I$, $S^\top JS = J$, $\det S = 1$, the congruence $S^\top GS = G_0$, and Theorem 7.2(1)–(3) are formalized without unproved placeholders (declarations `K_sq`, `S_symplectic`, `S_det`, `S_congruence`, `no_posdef`, `psd_degenerate`, `nondegenerate_indefinite`); SymPy 1.14.0, Python 3.14; the release tag `paper1-v1.1` fixes the commit. We do not claim formal verification of the operator-theoretic statements of Section 5, which remain at the form-identity level on $\mathcal{F}_{\mathrm{alg}}$.

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